This is your first homework assignment. I strongly encourage you to typeset it using the LaTeX template provided. It is due by Monday, January 13th.

The following question is on matrix multiplication.

(1) Suppose that $A \in \mathbb{R}^{m\times n}$, $\mathbf{v} \in \mathbb{R}^n$ and $B \in \mathbb{R}^{n\times r}$. Determine the number of floating point operations (*ie* additions, multiplications and divisions) required to compute $A\mathbf{v}$ and AB.

The following questions are on matrix and vector norms

(2) Let $\mathbf{v} = [1, 0, 5, -3]^\top$ and $\mathbf{w} = [-2, 4, 5, 1]^\top$.
   (a) Compute $\|\mathbf{v}\|_1$, $\|\mathbf{w}\|_3$ and $\|\mathbf{v}\|_\infty$.

   (b) Verify that $\langle \mathbf{v}, \mathbf{w}\rangle \leq \|\mathbf{v}\|_2\|\mathbf{w}\|_2$ by computing both sides.

(3) **This question will not be graded, so you may look up the answer in a textbook or online if you wish.** Show that $\|\mathbf{v}\|_p := \left(\sum_{i=1}^n |v_i|^p\right)^{1/p}$ is a norm for $1 \leq p \leq \infty$. We will call this norm "the p-norm" or the "ℓ_p norm" (pronounced "ell-p"). *Hint: you will need to use Hölder's inequality:*

$$\sum_{i=1} |v_i w_i| \leq \|\mathbf{v}\|_p\|\mathbf{w}\|_q \text{ for all } \mathbf{v}, \mathbf{w} \in \mathbb{R}^n \text{ and } q := \frac{1}{1-\frac{1}{p}} \text{ (equivalently: } \frac{1}{p} + \frac{1}{q} = 1\text{)}$$

(4) Prove that $\lim_{p\to\infty} \|\mathbf{v}\|_p = \|\mathbf{v}\|_\infty$. Recall that $\|\cdot\|_\infty$ is the max-norm defined as $\|\mathbf{v}\|_\infty := \max_i |v_i|$. *This is a hard question, so let me start you off:*

$$\begin{aligned}\|\mathbf{v}\|_\infty &= \left(\sum_{i=1}^n |v_i|^p\right)^{1/p} \qquad \text{Let } |v_{i^*}| = \max_i |v_i| \\ &= \left(|v_{i^*}|^p \left(1 + \sum_{i\neq i^*} \frac{|v_i|^p}{|v_{i^*}|^p}\right)\right)^{1/p}\end{aligned}$$

(5) Prove that $\|A\|_\infty = \max_{1\leq i\leq n} \sum_{j=1}^m |a_{ij}|$. *(Hint: you probably want to use the definition* $\|A\|_\infty = \max_{\|\mathbf{v}\|_\infty=1} \|A\mathbf{v}\|_\infty$*)*

(6) Suppose that $A \in \mathbb{R}^{m\times n}$, $\mathbf{b} \in \mathbb{R}^m$ and $\mathbf{x} \in \mathbb{R}^n$. Let $f(\mathbf{x}) = \|\mathbf{b} - A\mathbf{x}\|_2^2$. Show that $\nabla f(\mathbf{x}) = 2A^\top(A\mathbf{x} - \mathbf{b})$. *Hint: it is easiest to show this component wise. That is, show that the i-th component of the vector $\nabla f(\mathbf{x})$ is equal to the i-th component of the vector $2A^\top(A\mathbf{x} - \mathbf{b})$*

The following questions are on linear systems of equations

(7) Let $A = \begin{bmatrix} 0 & 6 & -1 \\ 3 & 5 & 2 \\ 1 & -2 & 4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix}$.
   (a) Find L, U and P such that $PA = LU$ and L is lower triangular with ones on the diagonal, U is upper triangular and P is a permutation matrix.

   (b) Use this LU decomposition to solve the linear system $A\mathbf{x} = \mathbf{b}$.

(8) Use the normal equations to solve the least squares problem $\mathbf{x}^* = \text{argmin}\|\mathbf{b} - A\mathbf{x}\|_2$ where $A = \begin{bmatrix} 1 & 5 \\ -3 & 2 \\ 4 & -7 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 4 \\ -3 \\ 2 \end{bmatrix}$